

Let $u: I \rightarrow \mathbb{R}$, where u is an open rectangle in $\mathbb{R}^2$.
Exact differential Eq. Suppose a function $u(x,y)$ has continuous partial derivatives on I , then

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy. \quad \text{--- (1)}$$

We have $u(x,y) = c$ (constant) iff $du = 0$.

Any first order & first degree diff eq can be written as
 $M(x,y) + N(x,y) \frac{dy}{dx} = 0$, or equivalently,

$$M dx + N dy = 0. \quad \text{--- (2)}$$

Let I be an open rectangle in $\mathbb{R}^2$ and $M \neq N$ be continuous on I . A diff. eq. (2) is called an exact diff. eq. if $\exists$ a function $u: I \rightarrow \mathbb{R}$, having continuous partial derivatives, s.t.
 $M dx + N dy = du = 0. \quad \text{--- (3)}$

If diff. eq. (2) is exact then $u = c$ is a solution, where c is a constant and u is as in Eq. (3).

Moreover, if diff. eq. (2) is exact then from Eqs. (1) and (3),
 $M = \frac{\partial u}{\partial x} \quad \text{and} \quad N = \frac{\partial u}{\partial y}.$

If M and N have continuous partial derivatives on I , then
 $\frac{\partial M}{\partial y} = \frac{\partial^2 u}{\partial y \partial x} \quad \text{and} \quad \frac{\partial N}{\partial x} = \frac{\partial^2 u}{\partial x \partial y}.$

By Mixed derivative theorem,

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}. \quad \text{(Necessary condition).}$$

Ex: show that the above condition is not sufficient.

Thus, $M dx + N dy = 0$ is exact iff $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$

Suppose Eq (2) is exact, that means $\exists u$ s.t.

$$\frac{\partial u}{\partial x} = M \quad \text{+} \quad \frac{\partial u}{\partial y} = N.$$

To find a solution of Eq. (2):

① Take the partial integration of $\partial u / \partial x = M$, this gives,

$$u = \int M dx + \phi(y), \text{ where}$$

ϕ is an arbitrary function of y .

(2) Then find $\partial u / \partial y$ and put it in $\partial u / \partial y = N$ to find ϕ .

(3) This gives, a solution $u = c$ of Eq. (2).

Ex. 1 $y dx + x dy = 0$.

Soln: $y dx + x dy = d(xy) = 0$ (exact)
 $\Rightarrow xy = c$.

Ex. 2. $(2x + y^2) + 2xy \frac{dy}{dx} = 0$

Sol: Here, $M = 2x + y^2$, $N = 2xy$.

$$\therefore \frac{\partial M}{\partial y} = 2y \quad \& \quad \frac{\partial N}{\partial x} = 2y$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}. \quad \text{Thus the given diff eq is exact.}$$

Thus there exists u s.t. $\frac{\partial u}{\partial x} = M$ & $\frac{\partial u}{\partial y} = N$.

$$\frac{\partial u}{\partial x} = 2x + y^2$$

Take the partial integration w.r.t. x

$$u = x^2 + y^2 x + \phi(y).$$

We have $\frac{\partial u}{\partial y} = N$.

$$\therefore 2y\cancel{x} + \phi'(y) = 2y\cancel{x} \Rightarrow \phi'(y) = 0$$

$$\Rightarrow \phi(y) = c.$$

$$\therefore u = x^2 + y^2 x + c.$$

$$\therefore \text{a solution is given by } u = c_1$$

$$\Rightarrow x^2 + y^2 x + c = c_1 \Rightarrow \boxed{x^2 + y^2 x = c'}$$

Ex $(y \cos x + 2xe^y) + (\sin x + x^2e^y - 1)y' = 0.$

Solⁿ: Here $M = y \cos x + 2xe^y$ & $N = \sin x + x^2e^y - 1$

$$\Rightarrow \frac{\partial M}{\partial y} = \cos x + 2xe^y, \quad \frac{\partial N}{\partial x} = \cos x + 2xe^y$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}. \quad \text{The given eq is exact.}$$

Therefore, $\exists u$ s.t. $\frac{\partial u}{\partial x} = M$ & $\frac{\partial u}{\partial y} = N.$

$$\therefore \frac{\partial u}{\partial x} = y \cos x + 2xe^y.$$

Take the partial integration w.r to x ,

$$u = y \sin x + 2e^y \frac{x^2}{2} + \phi(y).$$

We have $\frac{\partial u}{\partial y} = N.$

$$\Rightarrow \sin x + x^2 e^y + \phi'(y) = \sin x + x^2 e^y - 1$$

$$\Rightarrow \phi'(y) = -1.$$

$$\Rightarrow \phi(y) = -y + c.$$

$$\therefore u = y \sin x + e^y x^2 - y + c$$

$$\therefore \text{Solution: } \boxed{u = y \sin x + e^y x^2 - y = c'}$$

Reducible to exact: $M + N \frac{dy}{dx} = 0.$

Suppose it is not exact. If there exists a function $F(x, y)$ s.t. $FM + FN \frac{dy}{dx} = 0$ becomes exact.

If such a function F exists then we call it an integrating factor (IF). Thus, in this case, we have

$$(FM)dx + (FN)dy = 0$$

$$\text{and } \frac{\partial}{\partial y}(FM) = \frac{\partial}{\partial x}(FN).$$

$$\Rightarrow F \frac{\partial M}{\partial y} + M \frac{\partial F}{\partial y} = F \frac{\partial N}{\partial x} + N \frac{\partial F}{\partial x} \quad \text{--- (3)}$$

Case 1: F is a function of x only, that means $\frac{\partial F}{\partial y} = 0$.
Eq(3) reduces to

$$F \frac{\partial M}{\partial y} = F \frac{\partial N}{\partial x} + N \frac{\partial F}{\partial x}$$

$$\Rightarrow F \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = N \frac{\partial F}{\partial x} \Rightarrow \frac{1}{F} \frac{\partial F}{\partial x} = \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$$

$$\Rightarrow \ln F = \int \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) dx + \ln c$$

$$F = c e^{\int \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) dx}$$

Remark: You can ignore the constant of integration c .

Case 2: Similarly, if F is a function of y only, show that

$$F = c e^{\int \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dy}$$

Home work: Show that the equation $Mdx + Ndy = 0$ can be expressed in the form

$$\frac{1}{2} (Mx + Ny) d(\ln(Ny)) + \frac{1}{2} (Mx - Ny) d(\ln(\frac{x}{y})) = 0.$$

Hence, deduce that

Case 3: (i) if $Mx + Ny = 0$, then IF $F = \frac{1}{Mx - Ny}$ and $\frac{x}{y} = c$ is a solution,

Case 4: (ii) if $Mx - Ny = 0$, then IF $F = \frac{1}{Mx + Ny}$ and $xy = c$ is a solution.

Case 5: If M and N are homogeneous of the same degree and $Mx + Ny = 0$ then IF $F = \frac{1}{Mx + Ny}$.

Case 6: If $M = f_1(xy)y$, $N = f_2(xy)x$ and $Mx - Ny \neq 0$, then IF $F = \frac{1}{Mx - Ny}$.

Case-7: If $x^\alpha y^\beta (mydx + nx dy) = 0$ or

$$x^\alpha y^\beta (mydx + nx dy) + x^{\alpha'} y^{\beta'} (m'y dx + n'x dy) = 0$$

then

$$IF = x^h y^k \quad \text{for some } h \text{ and } k.$$

Ex. Solve $(2xy - 9y^2) dy + (2x^2 - 6xy) dx = 0$.

Sol: Here $M = 2xy - 9y^2$ and $N = 2x^2 - 6xy$ are homogeneous of the same degree (degree 2) and
 $Mx + Ny = 2x^2y - 9xy^2 + 2x^2y - 6xy^2$
 $= 4x^2y - 15xy^2 = xy(4x - 15y) \neq 0$.

$$\therefore IF = \frac{1}{xy(4x - 15y)}.$$

Multiply the given diff. eq by the above IF,

$$\frac{(2xy - 9y^2)}{xy(4x - 15y)} dy + \frac{(2x^2 - 6xy)}{xy(4x - 15y)} dx = 0.$$

$$\Rightarrow \frac{(2x - 9y)}{x(4x - 15y)} dy + \frac{(2x - 6y)}{y(4x - 15y)} dx = 0.$$

The above diff. eq is exact.

By the partial fraction, we can write the above diff. eq. as

$$\left(\frac{A_1}{x} + \frac{B_1}{4x - 15y} \right) dy + \left(\frac{A_2}{y} + \frac{B_2}{4x - 15y} \right) dx = 0,$$

$$\text{where } A_1(4x - 15y) + B_1x = 2x - 9y \quad \text{and}$$

$$A_2(4x - 15y) + B_2y = 2x - 6y.$$

$$\Rightarrow A_1 = 9/15 = 3/5 \quad \text{and} \quad B_1 = -2/5,$$

$$A_2 = 1/2 \quad \text{and} \quad B_2 = 3/2.$$

$$\text{Thus, } \left(\frac{3}{5x} - \frac{2}{5(4x - 15y)} \right) dy + \left(\frac{1}{2y} + \frac{3}{2(4x - 15y)} \right) dx = 0.$$

$$\text{Here, } M = \frac{3}{5x} - \frac{2}{5(4x - 15y)} \quad \text{and}$$

$$N = \frac{1}{2y} + \frac{3}{2(4x - 15y)}.$$

$$\therefore \frac{\partial u}{\partial x} = M = \frac{3}{5x} - \frac{2}{5(4x - 15y)} \quad \text{for some } u.$$

Integrate it partially wrt x ,

$$u = \frac{3}{5} \ln x - \frac{2}{5} \frac{\ln(4x-15y)}{4} + \phi(y).$$

Differentiate the above eq partially wrt y,

$$\frac{\partial u}{\partial y} = -\frac{1}{10} \frac{1}{(4x-15y)} (-15) + \phi'(y).$$

We have $\frac{\partial u}{\partial y} = N$, thus

$$\frac{3}{2(4x-15y)} + \phi'(y) = \frac{1}{2y} + \frac{3}{2(4x-15y)}.$$

$$\therefore \phi'(y) = \frac{1}{2y} \quad \text{and} \quad \phi(y) = \frac{1}{2} \ln y + \ln c.$$

$$\begin{aligned} \therefore u &= \frac{3}{5} \ln x - \frac{1}{10} \ln(4x-15y) + \frac{1}{2} \ln y + \ln c \\ &= \ln \left(\frac{c x^{3/5} y^{1/2}}{(4x-15y)^{1/10}} \right). \end{aligned}$$

$\therefore$ sol: $u = c_1$, c_1 is a constant,

$$\ln \left(\frac{c x^{3/5} y^{1/2}}{(4x-15y)^{1/10}} \right) = c_1$$

$$\text{or } x^{3/5} y^{1/2} = c_2 (4x-15y)^{1/10}$$

$$\text{or } x^6 y^5 = c_3 (4x-15y).$$

Ex Solve $(xy \sin xy + \cos xy) y dx + (xy \sin xy - \cos xy) x dy = 0$.

Sol: Here $M = (xy \sin xy + \cos xy) y$,

$N = (xy \sin xy - \cos xy) x$, and

$$\begin{aligned} Mx - Ny &= (xy \sin xy + \cos xy) xy - (xy \sin xy - \cos xy) xy \\ &= 2xy \cos xy \neq 0. \end{aligned}$$

$$\text{Thus, IF} = \frac{1}{Mx - Ny} = \frac{1}{2xy \cos xy}.$$

Exercise: Find the solution of the given diff. eq.

Ex: Solve $(y^2 + 2x^2y) dx + (2x^3 - xy) dy = 0$.

Sol: The above diff. eq can be written as

$$y(ydx - xdy) + x^2(2ydx + 2xdy) = 0.$$

Thus, IF = $x^h y^k$ for some h and k .
(Here $\alpha = 0, \beta = 1, \alpha' = 2, \beta' = 0$).

$$\therefore x^h y^k (y^2 + 2x^2y) dx + x^h y^k (2x^3 - xy) dy = 0$$

is exact.

$$\text{This gives, } \frac{\partial}{\partial y} (x^h y^k (y^2 + 2x^2y)) = \frac{\partial}{\partial x} (x^h y^k (2x^3 - xy)).$$

$$\Rightarrow \frac{\partial}{\partial y} (x^h y^{k+2} + 2x^{h+2} y^{k+1}) = \frac{\partial}{\partial x} (2x^{h+3} y^k - x^{h+1} y^{k+1})$$

$$\Rightarrow (k+2)x^h y^{k+1} + 2(h+1)x^{h+2} y^k = 2(h+3)x^{h+2} y^k - (h+1)x^h y^{k+1}$$

$$\Rightarrow k+2 = -(h+1) \text{ and } 2(h+1) = 2(h+3)$$

$$\Rightarrow h+k = -3 \text{ and } h-k = -2$$

$$\Rightarrow h = -5/2 \text{ and } k = -1/2$$

$$\therefore \text{IF} = x^h y^k = x^{-5/2} y^{-1/2} = \frac{1}{x^{5/2} y^{1/2}}.$$

Exercise: Find the sol. of the given diff. eq.